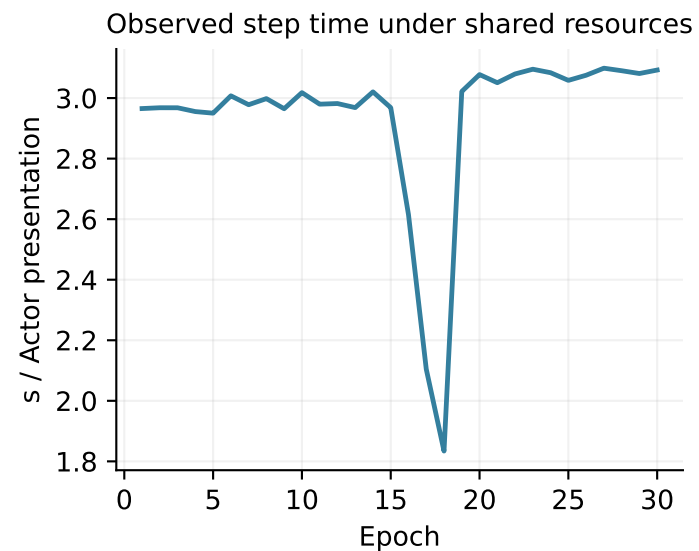
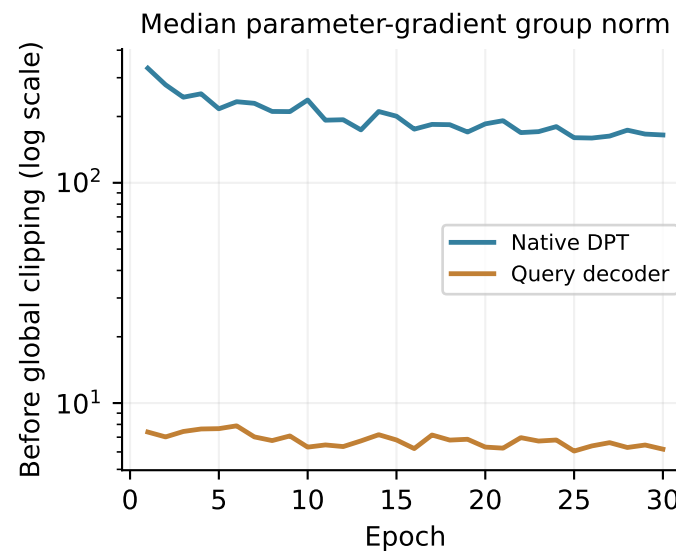
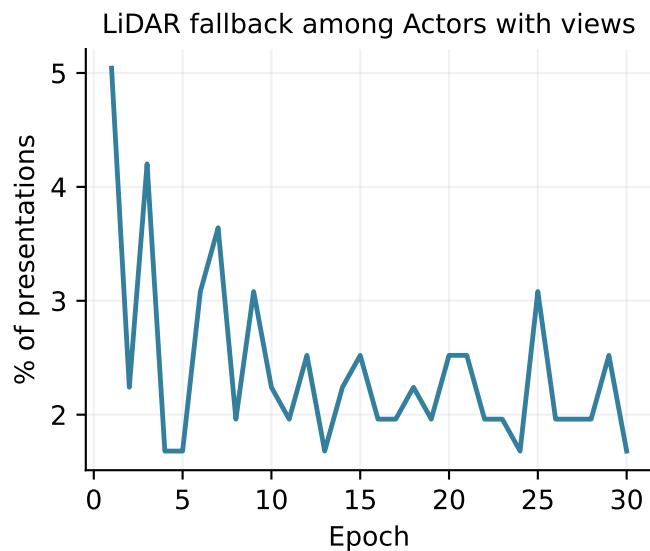
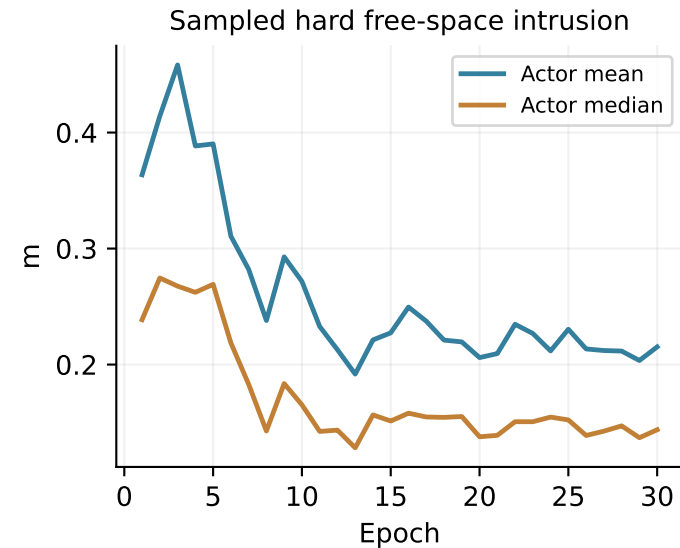
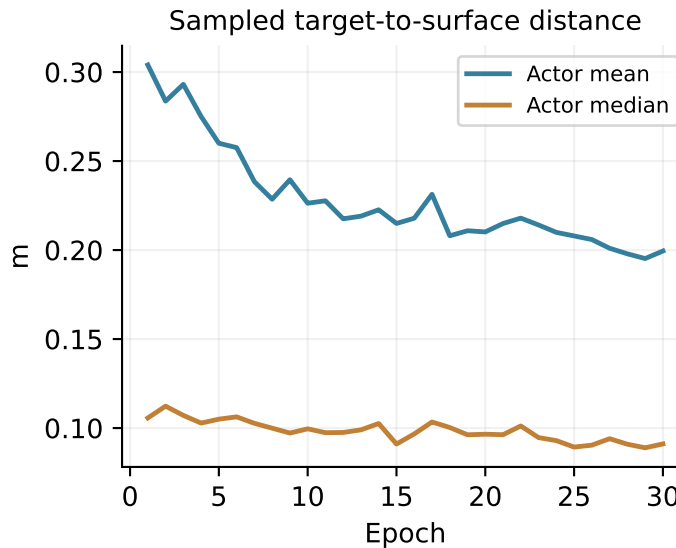
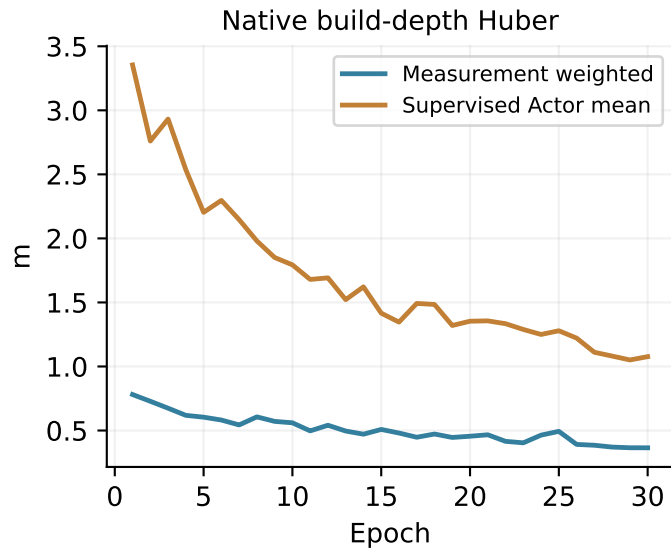


Joint native geometry + spatial queries: training process

371 fit presentations per recorded epoch; sampled observations vary, not paired generalization evidence.



No checkpoint resume. Actual optimizer updates: 11130. Skipped presentations: 0.

Native loss excludes unobserved Actors. Memory contention and concurrent work affect timing; loss reduction does not establish physical accuracy.